

ABSTRACT

Malicious attacks, malware, and ransomware families pose critical security issues to cybersecurity, and it may cause catastrophic damages to computer systems, data centers, web, and mobile applications across various industries and businesses. Traditional anti-ransomware systems struggle to fight against newly created sophisticated attacks. Therefore, state-of-the-art techniques like traditional and neural network-based architectures can be immensely utilized in the development of innovative ransomware solutions. In this paper, we present a feature selectionbased framework with adopting different machine learning algorithms including neural networkbased architectures to classify the security level for ransomware detection and prevention.

We applied multiple machine learning algorithms: Decision Tree (DT), Random Forest (RF), Naïve Bayes (NB), Logistic Regression (LR) as well as Neural Network (NN)-based classifiers on a selected number of features for ransomware classification. We performed all the experiments on one ransomware dataset to evaluate our proposed framework. The experimental results demonstrate that RF classifiers outperform other methods in terms of accuracy, F -beta, and precision scores.

INTRODUCTION

Malicious Applications or attacks, malware and ransomware families for instance, consistently endures to pose critical security issues to cybersecurity and it may cause catastrophic damages to computer systems, data centers, web, and mobile applications across various industries and businesses. Most ransomware is designed to block and prevent targeted victims from accessing computer data by applying an indestructible encrypting methodology that can be decrypted by the attacker itself solely. Removing the ransomware leads the victim to irreversible losses, as a result, victims are forced to pay according to the attacker's demands. Failure or denial to comply with the attacker's demand will lead to losing data permanently. With the help of modern technology, attackers are transforming conventional ransomware into emerging ransomware families which is more difficult in reversing a ransomware infection.

Ransomware is a sophisticated and variants threat affecting users worldwide that limits users from accessing their system or data, either by locking the system's screen or by encrypting and the users' files unless a ransom is paid. Two primary forms of ransomware based on attack approaches include locker ransomware that denies access to the computer or device and crypto ransomware that prevents access to files or data. After these attacks, it is incredibly difficult to revert without paying the extortion. Traditional ransomware detection techniques including event-based, statistical-based, and data-centric-based techniques are not adequate to combat. Therefore, implementing the highest level of optimal protection and security by adopting futuristic technology against such advanced malicious attacks should be imperative for the research community.

Novel technology, machine learning for instance in ransomware detection is a new research topic and can be immensely utilized in the development of innovative ransomware solutions. Employing the application of Machine Learning (ML) methodologies enables automatic detection of malware including ransomware through their dynamic behaviors and enhances security. Algorithms such as Decision Tree (DT), Random Forest (RF), Naïve Bayes(NB), Logistic Regression (LR), and Neural Network (NN)-based architectures have potential efficacy for ransomware classification and detection [9]. In this study, we conduct a comprehensive assessment and investigates the machine learning techniques for the classification of ransomware

1 Problem

Ransomware attacks are increasingly prevalent, posing significant threats to the integrity and availability of critical data across various sectors. Traditional cybersecurity measures are often inadequate in combating the evolving sophistication of ransomware, leading to severe financial and operational repercussions for organizations. The rapid development of new ransomware variants outpaces conventional detection systems, highlighting the need for innovative approaches to enhance detection and prevention capabilities.

This project aims to address the limitations of existing anti-ransomware solutions by leveraging advanced machine learning techniques for effective classification and detection of ransomware. By employing a feature selection-based framework and exploring various machine learning algorithms—including Decision Tree, Random Forest, Naïve Bayes, Logistic Regression, and Neural Networks—this research seeks to improve the accuracy, precision, and overall effectiveness of ransomware identification. The goal is to provide a robust mechanism that enhances cybersecurity resilience against emerging ransomware threats.

2 Research Question

Can a hybrid approach that combines multiple machine learning techniques improve ransomware detection rates compared to single-method approaches?

3 Purpose

The purpose of this project is to develop a robust framework for ransomware classification and detection using advanced machine learning algorithms. By employing a feature selection-based approach, the project aims to enhance the accuracy and effectiveness of identifying various ransomware families. Specifically, it will evaluate the performance of multiple algorithms, including Decision Tree, Random Forest, Naïve Bayes, Logistic Regression, and Neural Networks, to determine their efficacy in detecting ransomware. Additionally, the project seeks to identify critical features that significantly improve detection rates and propose innovative solutions that reduce the risks associated with ransomware attacks. Ultimately, this research aims to contribute valuable insights and tools to bolster cybersecurity measures across various industries.

4 Goals

Evaluate Machine Learning Algorithms: Assess the effectiveness of various machine learning algorithms (Decision Tree, Random Forest, Naïve Bayes, Logistic Regression, and Neural Networks) in accurately classifying and detecting different ransomware families.

Feature Selection Optimization: Identify and select the most relevant features that enhance the performance of machine learning models for ransomware detection.

Performance Comparison: Conduct a comprehensive analysis comparing the accuracy, precision, and F-beta scores of the different algorithms to determine the most effective approach for ransomware classification.

Develop a Detection Framework: Create a feature selection-based framework that integrates the best-performing machine learning algorithms for real-time ransomware detection and classification.

Contribute to Cybersecurity: Provide actionable insights and methodologies that can improve current cybersecurity practices, helping organizations mitigate risks associated with ransomware attacks.

Foster Future Research: Establish a foundation for further research in ransomware detection by documenting findings and suggesting potential areas for future exploration in machine learning applications for cybersecurity.

LITERATURE REVIEW

1 A Survey on Machine Learning Techniques for Cyber Security in the Last Decade

Pervasive growth and usage of the Internet and mobile applications have expanded cyberspace. The cyberspace has become more vulnerable to automated and prolonged cyberattacks. Cyber security techniques provide enhancements in security measures to detect and react against cyberattacks. The previously used security systems are no longer sufficient because cybercriminals are smart enough to evade conventional security systems. Conventional security systems lack efficiency in detecting previously unseen and polymorphic security attacks. Machine learning (ML) techniques are playing a vital role in numerous applications of cyber security.

2 Ransomware-Threats, Vulnerabilities And Recommendations

Attack methodologies transform with the transforming dynamics of technology. Consequently it becomes imperative that individuals and organization implement the highest levels of security within their devices and infrastructure for optimal protection against these rapidly evolving attacks. Ransomware is one such attack that never fails to surprise in terms of its ability to identify vulnerabilities and loopholes in technology. This paper discusses the categories of ransomware, its common attack vectors and provides a threat landscape with the aim to highlight the true potential and destructive nature of such malware based attacks. In this paper we also present the most current ransomware attack that is still a potential threat and also provide recommendations and strategies for prevention and protection against these attacks. A novel solution is also discussed that could be further worked upon in the future by other researchers and vendors of security devices.

Analysis of Machine Learning Techniques for Ransomware Detection

In parallel with the increasing growth of the Internet and computer networks, the number of malwares has been increasing every day. Today, one of the newest attacks and the biggest threats in cybersecurity is ransomware. The effectiveness of applying machine learning techniques for malware detection has been explored in much scientific research, however, there is few studies focused on machine learning-based ransomware detection. In this paper, the effectiveness of ransomware detection using machine learning methods applied to CICAndMal2017 dataset is examined in two experiments. First, the classifiers are trained on a single dataset containing different types of ransomware. Second, different classifiers are trained on datasets of 10 ransomware families distinctly.

Our findings imply that in both experiments random forest outperforms other tested classifiers and the performance of the classifiers are not changed significantly when they are trained on each family distinctly. Therefore, the random forest classification method is very effective in ransomware detection.

A Study on the Evolution of Ransomware Detection Using Machine Learning and Deep Learning Techniques

This survey investigates the contributions of research into the detection of ransomware malware using machine learning and deep learning algorithms. The main motivations for this study are the destructive nature of ransomware, the difficulty of reversing a ransomware infection, and how important it is to detect it before infecting a system. Machine learning is coming to the forefront of combatting ransomware, so we attempted to identify weaknesses in machine learning approaches and how they can be strengthened. The threat posed by ransomware is exceptionally high, with new variants and families continually being found on the internet and dark web. Recovering from ransomware infections is difficult, given the nature of the encryption schemes used by them. The increase in the use of artificial intelligence also coincides with this boom in ransomware. The exploration into machine learning and deep learning approaches when it comes to detecting ransomware poses high interest because machine learning and deep learning can detect zero-day threats. These techniques can generate predictive models that can learn the behaviour of ransomware and use this knowledge to detect variants and families which have not yet been seen.

In this survey, we review prominent research studies which all showcase a machine learning or deep learning approach when detecting ransomware malware.

These studies were chosen based on the number of citations they had by other research. We carried out experiments to investigate how the discussed research studies are impacted by malware evolution. We also explored the new directions of ransomware and how we expect it to evolve in the coming years, such as expansion into IoT (Internet of Things), with IoT being integrated more into infrastructures and into homes.

Signature-Based Malware Detection Using Sequences of N-grams

Signature-based antivirus, as a form of malware detection technique, has the ability to detect and remove any known malware by simply identifying the signature of any malicious file present in the database. The growth in malware has been considerable and become an important problem that threatens the security of any computer system. The focus of our study is on one set of malware family, i.e. the Brontok worms, because they have long been a huge burden to most Windows-based user platforms because of some unique qualities. A prototype of the antivirus was able to scan files

and accurately detect any traces of the Brontok malware signatures in the scanned files. In this study, we developed a detection model by extracting the signatures of the Brontok worms and used an n-gram technique to break down the signatures. This process made it easier to remove redundancies between the signatures of the different types of Brontok malware, which was used in this study and could accurately differentiate between the signatures of both malicious and normal files. During the experiment, we successfully detected the presence of Brontok worms while correctly identifying the benign ones. The techniques employed in the experiment provided some insight on creating a good signature-based detector, which could be used to create a more credible solution that eliminates any threats of old malware, which may resurface in the future.

METHODOLOGY

MACHINE LEARNING ALGORITHMS

Machine learning algorithms are computational models that enable machines, particularly computers, to learn from data and make predictions or decisions without being explicitly programmed. These algorithms utilize statistical techniques to recognize patterns, learn from experience, and improve their performance over time as they are exposed to more data.

RANDOM FOREST ALGORITHM

The Random Forest Classifier is an ensemble learning algorithm that belongs to the family of decision tree-based algorithms. It is widely used for both classification and regression tasks. Random Forests build multiple decision trees during training and merge their predictions to provide a more accurate and stable result.

The random forest algorithm is made up of a collection of decision trees, and each tree in the ensemble is comprised of a data sample drawn from a training set with replacement, called the bootstrap sample. Of that training sample, one-third of it is set aside as test data, known as the out-of-bag (oob) sample, which we'll come back to later. Another instance of randomness is then injected through feature bagging, adding more diversity to the dataset and reducing the correlation among decision trees. Depending on the type of problem, the determination of the prediction will vary. For a regression task, the individual decision trees will be averaged, and for a classification task, a majority vote—i.e. the most frequent categorical variable—will yield the predicted class. Finally, the oob sample is then used for cross-validation, finalizing that prediction

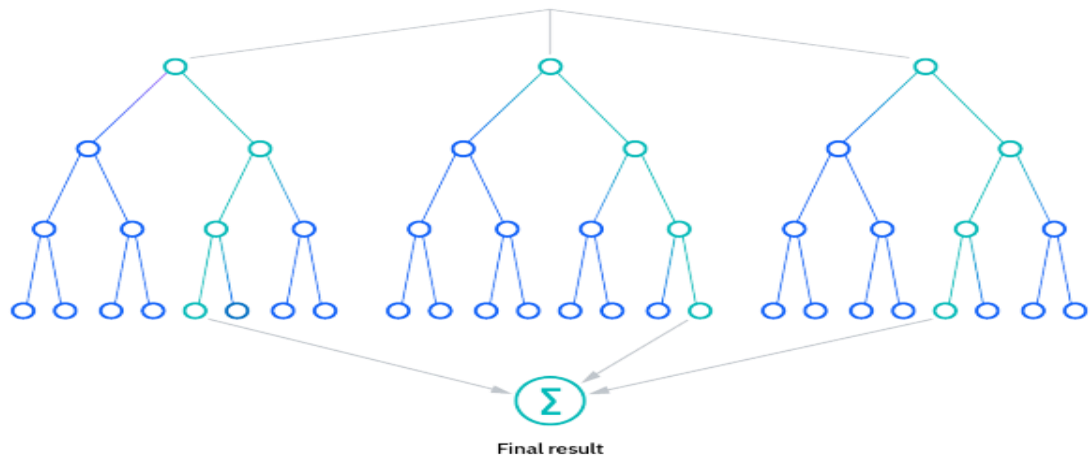


Figure Decision Tree

The key steps involved in the Random Forest algorithms:

- **RANDOM SAMPLING:** Randomly select subsets of the training data (with replacement) to train each tree. This process is known as bootstrap sampling.
- **RANDOM FEATURE SELECTION:** For each tree, randomly select a subset of features at each split point during the construction of the tree. This helps in creating diverse trees and reduces overfitting.
- **DECISION TREE CONSTRUCTION:** Construct a decision tree for each subset of data using the selected features. The tree is grown until a stopping criterion is met, such as a maximum depth or the number of data points in a leaf.
- **VOTING(CLASSIFICATION) OR AVERAGING(EGRESSION):** For a new input, each tree in the forest predicts an outcome. In the case of classification, the class that receives the majority of the votes is the final prediction. For regression, the predictions are averaged to obtain the final result.

The advantages of Random Forest includes:

- **ENSEMBLE LEARNING:** Combining the predictions of multiple models often leads to better overall performance and generalization.
- **REDUCED OVERFITTING:** The randomness in feature selection and data sampling helps to reduce overfitting compared to individual decision trees.

- **HIGH ACCURACY:** Random Forest tends to provide high accuracy in both classification and regression tasks.
- **ROBUSTNESS:** It is robust to outliers and noise in the data.
- **IMPLICIT FEATURE IMPORTANCE:** Random Forest provides a measure of feature importance, indicating which features are more influential in making predictions.

How does Random Forest algorithm work?

Random Forest works in two-phase first is to create the random forest by combining N decision tree, and second is to make predictions for each tree created in the first phase.

The Working process can be explained in the below steps and diagram:

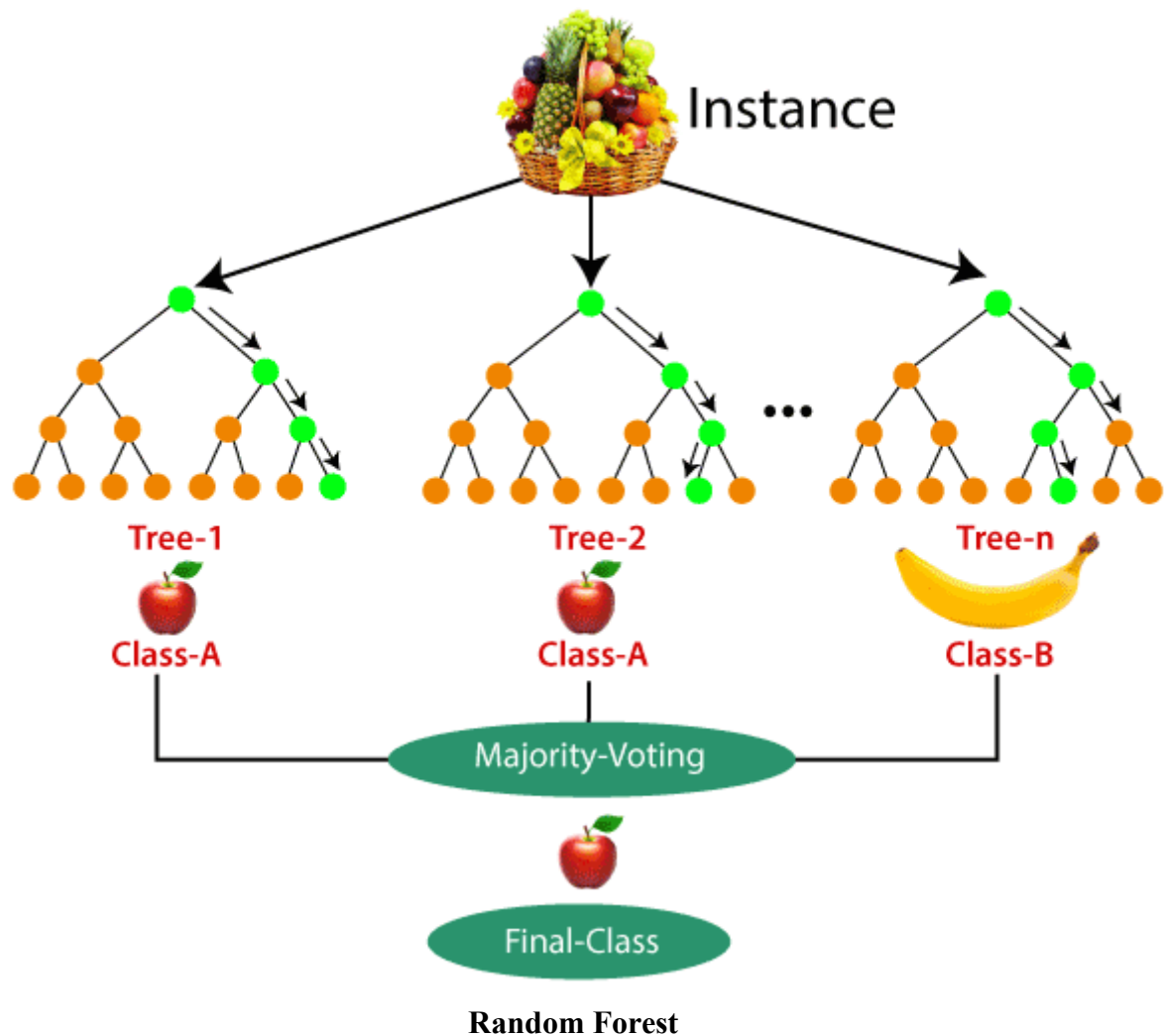
Step-1: Select random K data points from the training set.

Step-2: Build the decision trees associated with the selected data points (Subsets).

Step-3: Choose the number N for decision trees that you want to build.

Step-4: Repeat Step 1 & 2.

Step-5: For new data points, find the predictions of each decision tree, and assign the new data points to the category that wins the majority votes.



Decision Tree Classifier

The Decision Tree classifier is a powerful and interpretable machine learning algorithm that makes decisions based on the features of the input data. It works by recursively splitting the dataset into subsets based on feature values, resulting in a tree-like structure where each internal node represents a feature, each branch represents a decision rule, and each leaf node represents the final output (class label).

The decision-making process involves selecting the feature that provides the best separation of the classes at each node. This is often measured using metrics such as Gini impurity, which quantifies how often a randomly chosen element would be incorrectly labeled if it were randomly labeled according to the distribution of labels in the subset. Alternatively, entropy can also be used to measure the uncertainty in the dataset. The tree continues to split until a stopping criterion is reached, such as a maximum depth, a minimum number of samples in a node, or no further improvement in splitting. One of the key advantages of Decision Trees is their interpretability. The

clear structure allows stakeholders to understand how decisions are made, which is essential in fields like cybersecurity where transparency is crucial. Additionally, Decision Trees can handle both categorical and numerical data, making them versatile for various applications.

However, Decision Trees have a tendency to overfit, especially when the tree is deep or the dataset is noisy. Overfitting occurs when the model learns to capture noise in the training data rather than the underlying pattern, resulting in poor generalization to new data. To address this, techniques such as pruning (removing parts of the tree that do not provide significant power) or limiting the maximum depth can be applied. Despite these drawbacks, Decision Trees serve as the foundation for more complex ensemble methods like Random Forest and Gradient Boosting, making them essential in the machine learning toolkit.



Decision Tree

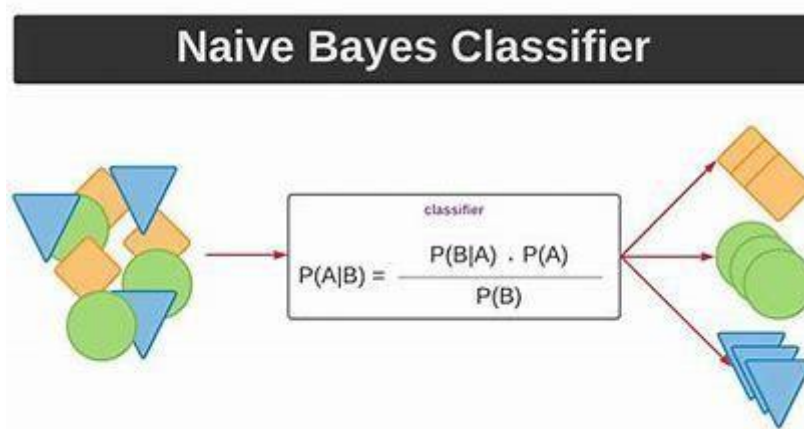
Naive Bayes Classifier

Naive Bayes is a family of probabilistic classifiers based on Bayes' theorem, which calculates the probability of a class based on prior knowledge of conditions related to the class. The model assumes that all features are independent given the class label, which simplifies the calculations involved in determining the probabilities.

There are several variations of Naive Bayes classifiers, including Gaussian Naive Bayes (for continuous data), Multinomial Naive Bayes (for discrete counts), and Bernoulli Naive Bayes (for binary/boolean features). The model estimates the conditional probability of each feature given the class label and uses these probabilities to make predictions by applying Bayes' theorem.

Naive Bayes classifiers are particularly well-suited for text classification tasks, such as spam detection or sentiment analysis, where the independence assumption holds reasonably well. The algorithm is efficient and can handle large datasets effectively, making it a popular choice for realtime applications.

Despite its strengths, the Naive Bayes classifier has limitations. The independence assumption can lead to underperformance when features are correlated, as the model fails to capture the interactions between them. Additionally, Naive Bayes may struggle with rare events, as the probabilities can be skewed, leading to inaccurate predictions. Techniques like Laplace smoothing can be applied to mitigate the impact of zero probabilities, enhancing the model's robustness. Overall, Naive Bayes is valued for its simplicity, speed, and effectiveness in various applications, particularly in scenarios where interpretability and computational efficiency are crucial.



Naïve Bayes Classifier

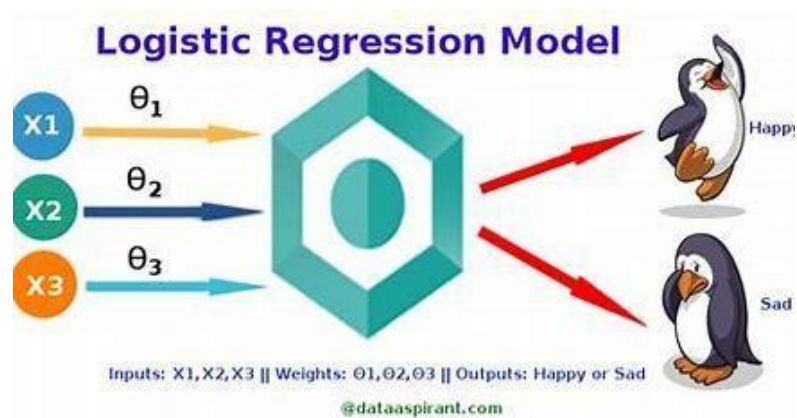
Logistic Regression

Logistic Regression is a statistical method used for binary classification that models the probability of a class based on one or more predictor variables. It operates by fitting a linear equation to the input features and applying the logistic function (sigmoid function) to map the predicted values to probabilities between 0 and 1. The logistic function is defined as $P(Y=1|X) = \frac{1}{1 + e^{-z}}$, where z is a linear combination of the input features. The coefficients of the model are estimated using maximum likelihood estimation (MLE), which seeks to find the parameter values that maximize the likelihood of the observed data given the model.

One of the key advantages of Logistic Regression is its interpretability. The coefficients obtained from the model provide insights into the relationship between each feature and the outcome, allowing stakeholders to understand which variables are driving the predictions. This

interpretability is particularly valuable in applications like medical diagnosis or credit scoring, where understanding the factors influencing decisions is crucial.

However, Logistic Regression assumes a linear relationship between the features and the log-odds of the outcome, which may not hold true in more complex datasets. It is also sensitive to outliers, which can disproportionately influence the model's predictions. Moreover, the model requires a relatively large sample size for stable estimates, especially when dealing with multiple predictor variables. Despite its limitations, Logistic Regression remains a widely used algorithm due to its simplicity, efficiency, and effectiveness in various classification tasks, including ransomware detection, where it can help identify malicious behavior based on specific features.



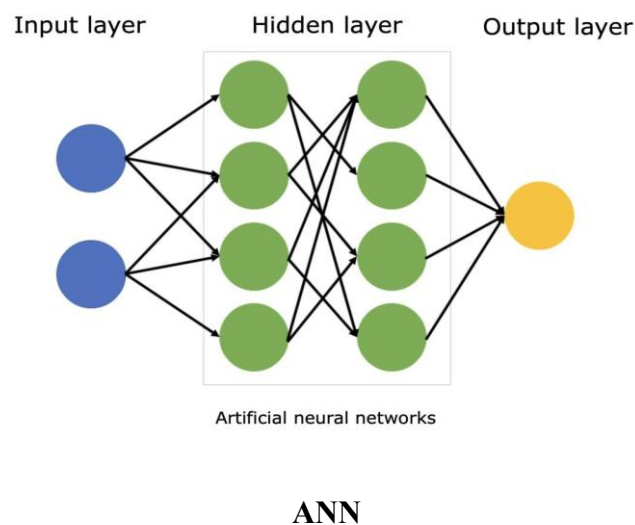
Logistic Regression Model

Neural Network

Neural Networks are computational models inspired by the human brain, consisting of interconnected nodes (neurons) organized in layers. Each neuron receives inputs, applies weights to them, and passes the result through an activation function to produce an output. The architecture typically includes an input layer, one or more hidden layers, and an output layer. During the training process, the neural network learns the optimal weights for each connection through a method called backpropagation. In backpropagation, the model's predictions are compared to the actual outputs, and the error is propagated back through the network to adjust the weights based on the gradient of the loss function. This iterative process allows the model to minimize the error and improve its accuracy over time.

Neural Networks excel in handling large datasets and can capture intricate patterns due to their ability to learn complex, nonlinear relationships. They are particularly effective in tasks involving unstructured data, such as image recognition, natural language processing, and audio

classification. However, Neural Networks require significant computational resources and can be prone to overfitting, especially with small datasets. To mitigate overfitting, techniques such as dropout (randomly disabling neurons during training) and regularization (adding a penalty for complexity) can be employed. The design of the network, including the number of layers, the number of neurons per layer, and the choice of activation functions, can significantly influence performance. In ransomware detection, Neural Networks can learn complex feature interactions that traditional algorithms might miss, enhancing classification accuracy. They provide the flexibility and scalability needed for modern machine learning applications, making them a powerful tool in the fight against cyber threats.



TOOLS/ HARDWARE & SOFTWARE REQUIREMENTS

HTML and JavaScript: The HTML structure defines the login page content, layout, and interactive elements. JavaScript provides dynamic functionality, including interactions and input validation.

reCAPTCHA:

This page references Google reCAPTCHA (<https://www.gstatic.com/recaptcha>) to prevent automated bots from accessing the login system, providing an added layer of security.

Wordtune: A script for Wordtune is installed on the page, which might assist in text-related enhancements.

Tailwind CSS: Tailwind CSS framework is applied for utility-based styling, allowing modular design and rapid customization.

Font Awesome Icons: Several icon classes (svg-inline--fa) suggest the use of Font Awesome for icons, aiding in visual cues for users on the login page.

HARDWARE-REQUIRMENTS:

- System : i3 or above.
- Ram : 4 GB.
- Hard Disk : 40 GB

SOFTWARE REQUIRMENTS :

- Operating system : Windows8 or Above.
- Coding Language : python

CONCLUSION

Malware including ransomware is increasingly posing a serious security threat to financial institutions, businesses, and individuals. It is essential to develop an automatic system to effectively classify and detect ransomware and reduce the risk of malicious activities. In this paper, we presented a feature selection-based novel framework, adopted different machine learning algorithms including neural network-based classifiers for effective ransomware classification and detection. We applied the framework with all the experiments on a ransomware dataset and evaluated the models' performance by a robust comparative analysis among DT, RF, NB, LR, and NN classifiers. The experimental results demonstrate that the Random Forest classifier outperformed other classifiers by achieving the highest accuracy, F-beta, and precision scores with reasonable consistency in the 10fold cross-validation res

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